

5CS045 Server Capacity for Exams

Sep 30, 2026 · @Sarayu

Ask IT for **32 GB of memory and 16 vCPUs, both reserved**. A 400-student exam needs about 22 GB of memory and the server has 11 GB. The 443 GB disk is already enough.

Today

The server is nearly idle: 8 students have registered so far and use 21 MB between them. The ask is sized for the exam peak, not for today's use.

	Today
Processor	8 vCPUs, load 0.06
Memory	11 GB, 10 GB of it available
Disk	443 GB, 15 GB used (4%)
Students	8 registered, 21 MB of files, databases empty

Load tests

Tests on 23 and 24 Sep with 800 test accounts show the server now handles a whole cohort logging in and uploading over two minutes. They also found five problems: four are fixed, and the fifth now recovers on its own.

Where it stands now (24 Sep)	Result
800 students log in over 2 minutes	All 800 got in; median login 0.2 s; CPU peaked at 50%
800 students upload 5 MB each over 2 minutes	All 800 succeeded; CPU peaked at 61%
800 students log in within the same second	About 300 got in, then the login service stalled; it now restarts itself within 2 minutes

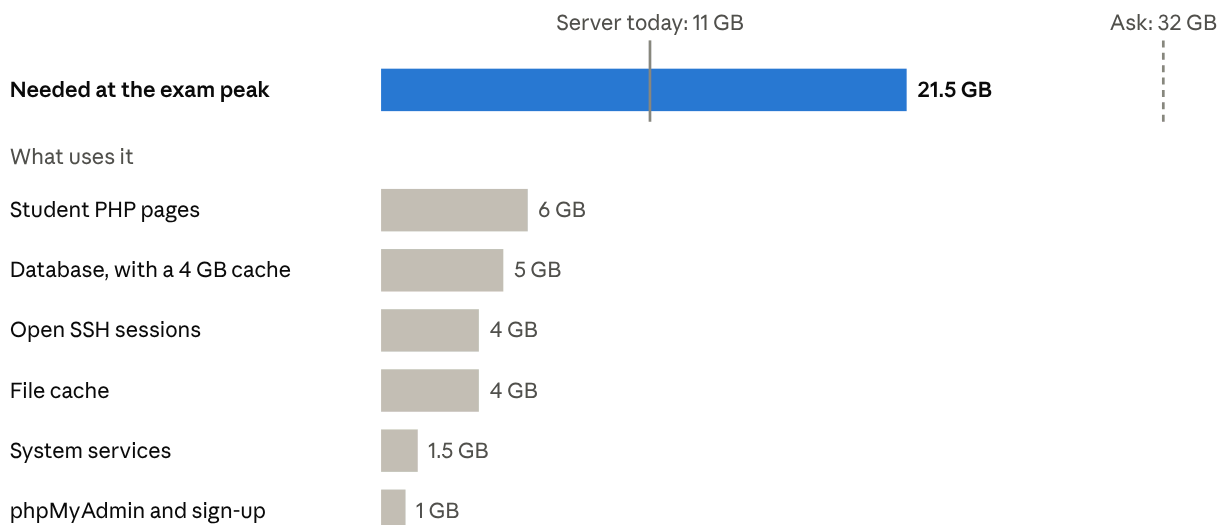
Date	What the tests found	What we changed
24 Sep	400 and 800 logins: 250 and 501 terminals timed out, the load average hit 125 and about 1 GB went to swap	Stopped a background process that started for every student at login
23 Sep	400 logins: 171 students got no CPU or memory limits	Raised the system's limit on login sessions
23 Sep	60 logins at once: 25 were refused	Raised SSH's limit on logins in progress
23 Sep	900 student sites: pages failed after about 56 busy PHP processes	Raised PHP's open-file limit

The tests covered SSH logins and uploads only. PHP pages and the database were not load-tested, so the exam estimate below is built from per-student measurements.

Exam peak

400 students at once need about 22 GB of memory; the server has 11 GB

Memory at the exam peak, and what uses it



Measured on a test copy of the server and scaled to 400 students; server size read on 30 Sep 2026

Each student's share was measured on a test copy of this server: about 8 MB per open SSH session, 7 to 12 MB per PHP page process and 28 MB per phpMyAdmin process.

The ask

Memory and processor are what an exam runs short of; the disk already covers the cohort.

	Ask	What it buys
Memory	32 GB, reserved	400 students at once with about 30% to spare, or 800 at the limit
Processor	16 vCPUs, reserved	The final-minute upload rush takes about 5 cores on its own. With pages, phpMyAdmin and the database on top, 8 cores run flat out; 16 run at about half
Disk	Keep the 443 GB; confirm it is SSD	Room for about 670 students at their full 600 MB allowance
Network	1 Gbps to the labs	400 final uploads of 5 MB in one minute need about 270 Mbit/s
Backups	A snapshot before and after each exam, and daily backups	Exam submissions cannot be lost

Reserved means the host does not share this CPU and memory with other virtual machines. Without that, an exam gets whatever is left over.

Our side

Three settings would still choke an exam on bigger hardware. Raising them needs the new memory, so we change them once the upgrade is in.

Setting	Now	After the upgrade
PHP processes for phpMyAdmin and sign-up	5	40
Database connections	151	500
Database cache	128 MB	4 GB

We will also stagger lab finish times by 2 to 3 minutes, so a whole cohort never logs in within the same second.